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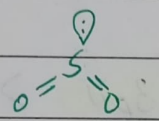
CHEMICAL BONDING

* $\left(\frac{\text{C. Atom } (v.e^-) + \text{Surr. atoms } (v.e^-) + (-\text{ve charge}) - (+\text{ve charge})}{8}\right)$

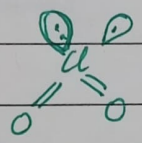
↳ Quotient = 8. Pairs
Rem = L.P

(H as a side atom → valence e⁻s to be taken 7)

eg. SO_2 ⇒ $8\overline{)6 + 12 + 0 = 18}$ → (2) → B.P
16
(2) → 2/2 = 1 L.P.

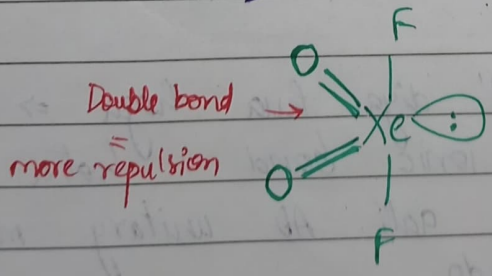


eg. → ClO_2 ⇒ $\frac{7 + 12}{8}$ = 2 BP + 1 LP + 1 e⁻

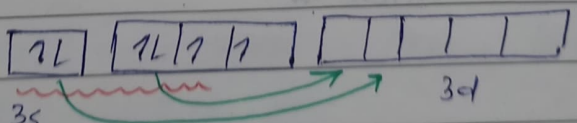


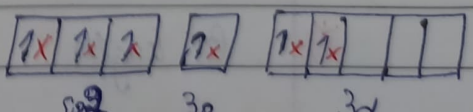
* Axial < Equitorial (less repulsion in eq.)
∴ LP is kept in eq. and not axial.

Ily in XeO_2F_2

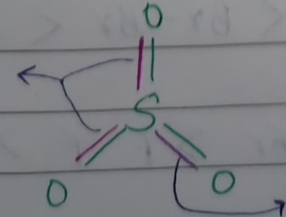


* SO_2 →



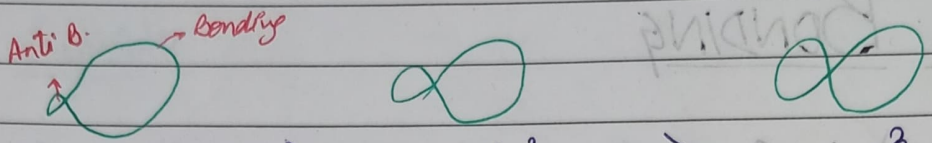


3d of S and 2p of O



→ $2p - 3p_{(S)}$ 'O'

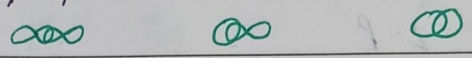
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* 

[sp > sp^2 > sp^3] B. strength

* B. strength (σ)

$p-p$ > $s-p$ > $s-s$

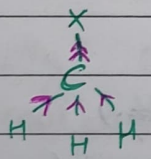


($\sigma > \pi$) strength

*// $2p\pi - 2p\pi$ > $2p\pi - 3d\pi$ > $2p\pi - 3p\pi$ > $3p\pi - 3p\pi$

Dipole Moment

$\vec{\mu} = q \times d$



★ $CH_3Cl > CH_3F > CH_3Br > CH_3I$

$1\,D = 3.33 \times 10^{-30}\ C\,m$

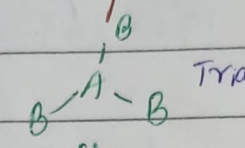
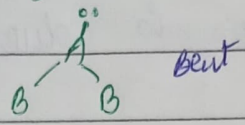
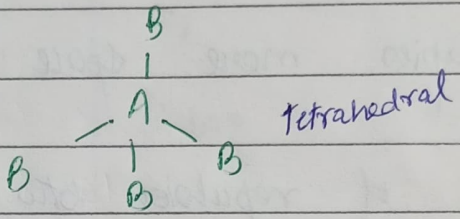
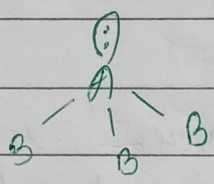
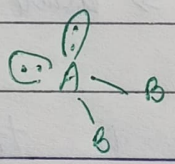
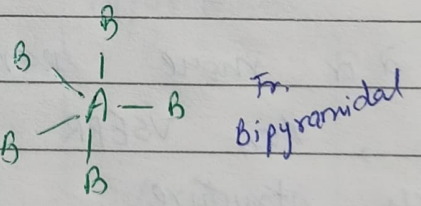
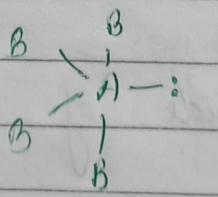
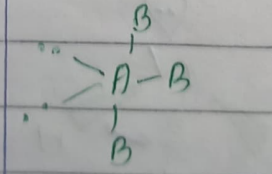
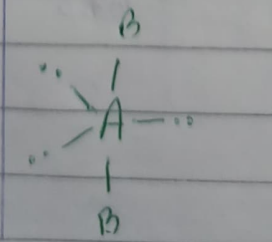
$1\,A^\circ = 10^{-10}\ m$

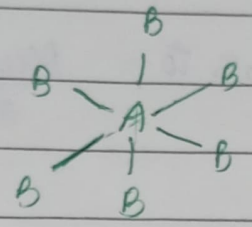
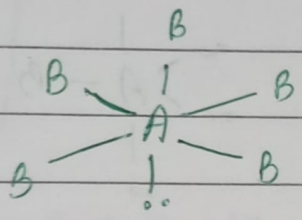
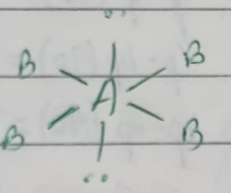
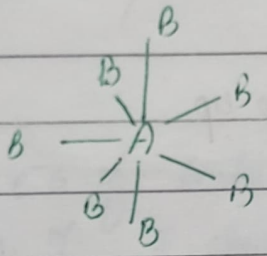
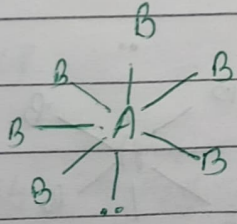
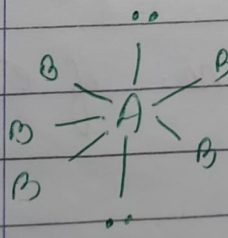
Ques mein 'μ' aur 'd' diya hua hoga ⇒ usse 'q' nikaal do. Agar 100% ionic bond hota toh q ki value 1.6×10^{-19} C aati. Ab unitary method se % ionic char. nikal do.

*// B. length → $F-F < Cl-Cl < Br-Br < I-I$

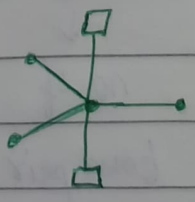
B. strength → $Cl-Cl > Br-Br > F-F > I-I$

Central atom w/ LPs.

S(LP+BL)	Molecule Type	Geometry	Shape	Ideal Bond angle
3	AB_3	Triangular	 Triangular	120°
	AB_2L		 Bent	
4	AB_4	Tetrahedral	 Tetrahedral	109°28'
	AB_3L		 Pyramidal	
	AB_2L_2		 Bent	
5	AB_5	Trigonal Bipyramidal	 Tr. Bipyramidal	
	AB_4L		 See saw	120°, 90°, 180°
	AB_3L_2		 T-shaped	
	AB_2L_3		 linear	

	AB_6		 Sq. b.Py.	90°, 180°
6		octahedral / Square B-py.		
	AB_5L		 square pyramidal	
	AB_4L_2		 square planar	
	AB_7		 Pentagonal bipyramidal	72°, 90°, 180°
		Pentagonal bipyramidal		
7	AB_6L		 Pentagonal pyramidal	
	AB_5L_2		 Pentagonal planar	

Equitorial position ⇒ round the middle (○)
Axial / apical position ⇒ up / down (□)

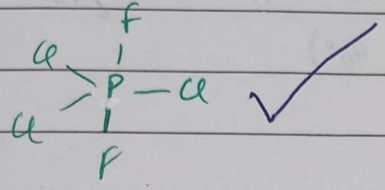


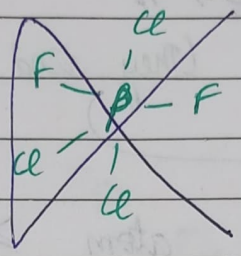
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Hybridisation Type	Geometry	Participating Orbitals
sp	Linear	S + any P
sp^2	Triangular	S + any 2 P
sp^3	Tetrahedral	$S + P_x + P_y + P_z$
sp^3d	TBP	$S + P_x + P_y + P_z + P_z^2$
	Square Pyramidal	$S + P_x + P_y + P_z + d_{x^2-y^2}$
sp^3d^2	Octahedral	$nS + nP_x + nP_z + nP_y + nd_{x^2-y^2} + nd_{z^2}$
d^2sp^3	"	$nS + nP_x + nP_y + nP_z + (n-1)d_{x^2-y^2} + (n-1)d_{z^2}$
d^3s	Tetrahedral	$ns + (n-1)d_{xy} + (n-1)d_{xz} + (n-1)d_{yz}$
dsp^2 / sp^2d	Square Planar	$(n-1)d_{x^2-y^2} + ns + np_x + np_y$ $nd_{x^2-y^2}$ ✱
sp^3d^3	PBP	$ns + np_x + np_y + np_z + nd_{x^2-y^2} + nd_{xy} + d_{z^2}$

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* $sp^3d \Rightarrow sp^2 + pd$ ↳ $3d_{z^2}$





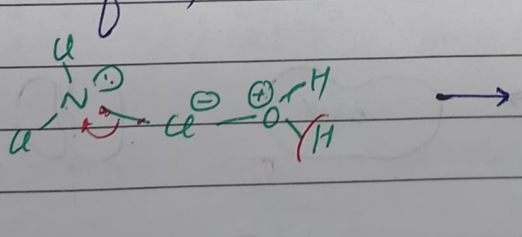
PF_2Cl_3 ??

{ More eN atom will overlap with pd and not sp^2 ⇒ taaki ub zyada kheech paaye

** Lewis Acid strength ⇒ XXX $BF_3 > BCl_3 > BBr_3 > BI_3$ XXX

(-I effect and back bonding) ✓ $BF_3 < BCl_3 < BBr_3 < BI_3$

* NCl_3 goes under hydrolysis even after having no empty d orbital on N. (Attack on empty d obbital of Cl)



↳ NH_3

{ In CCl_4, attack on empty 3d of Cl is not possible due to -δ charge on Cl }

* Induced Dipole - Induced Dipole AKA London Dispersion forces are the weakest IMF.

↓

** LDF ∝ M.M ∝ Surface Area

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* H-Bonding → $H_2O > HF > NH_3$

** $PH_3 \Rightarrow 93.8°$ $H_2S \Rightarrow 92°$ $H_2Se \Rightarrow 91°$ $H_2Te \Rightarrow 89.5°$
$H_2S \Rightarrow 92°$ (They tend to 90°)
$NO_2^- \Rightarrow 134°$

Dragol's rule → Central atom ≥ 3rd period, L·P goes in s orbital and bonding occurs with p orbitals making the B. angle ≈ 90°.

MOT

2 atomic orbital = 2 mol. orbital → Bonding, Anti Bonding

$$B.O = \frac{N_B - N_A}{2}$$

{ BO ↑ ⇒ B.S ↑ ⇒ B.L ↓ } Agar BO =, then more e^- in Anti Bonding will be unstable

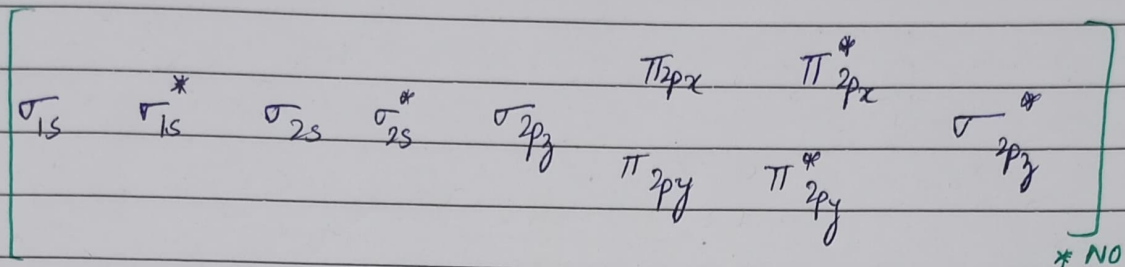
① s + s → Bond. σ_{1s} , A.B σ^*_{1s}

② p_z + p_z → B. σ_{p_z} + A.B $\sigma^*_{2p_z}$

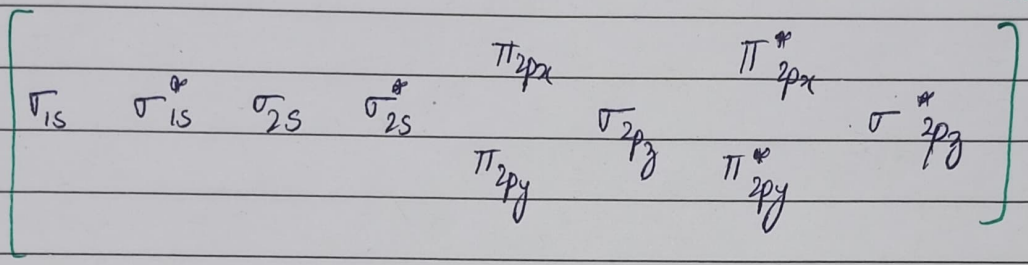
③ p_x/p_z + p_x/p_z → B π_{2p_x} + A.B $\pi^*_{2p_x}$

** s-p mixing (≤ 14)
↳ { σ_{2s} ↓ σ_{2p_z} ↑ ; σ^{**}_{2s} ↓ $\sigma^{**}_{2p_z}$ ↑ } energy

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* $NO^+ \rightarrow 14\ e^-s$
but $NO \rightarrow 15$ toh NO ke config se ek e^- nikala jaega

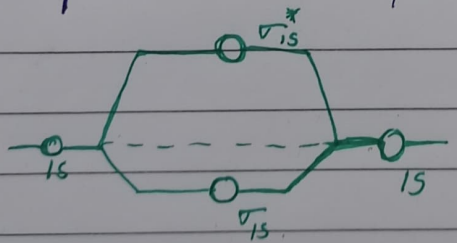


* $C_2 \rightarrow$ B.O $\rightarrow 2$
$\rightarrow$ 0 σ, 2 π

e^- =>	8	9	10	11	12	13	14	15	16	17	18	19	20
	0	0.5	1	1.5	2	2.5	3	2.5	2	1.5	1	0.5	0

HOMO → Highest occupied M.O
LOMO → Lowest unoccupied M.O

Unpaired e^- => paramagnetic paired e^- => diamagnetic



σ^*_{1s} ki zyada badhti hai
σ_{1s} ki kamm kamm hoti hai
energy.

σ - Bonding and π^* - anti Bonding M.O] → Gerade → symm. about centre
σ^* - Anti B. and π - Bonding M.O → Ungerade → opp. sign of wave about centre.